Leela Krishna Koppolu

leelakrishnakoppolu@gmail.com | +1 (682) 583-5890 | Dallas, TX (Open to Relocate)
linkedin.com/in/leela-krishna-klk | github.com/KoppoluLeelaKrishna

SUMMARY:

AI Data Engineer with 4+ years of experience building large-scale data platforms, ETL/ELT pipelines, and ML/AI data infrastructure across banking, retail, and enterprise. Skilled in AWS (S3, Glue, Lambda, EMR, Redshift, Kinesis, SageMaker, Bedrock, EKS), Snowflake, Databricks, Spark/PySpark, Airflow, Kafka, dbt, and modern AI tooling (LangChain, vector databases, RAG pipelines). Experienced in dimensional modeling, streaming ingestion, feature engineering for ML, data-quality engineering, and Spark/SQL tuning that reduces runtime and warehouse cost.

EDUCATION

Master of Science, Computer Science *Jan 2023 – Dec 2024*
The University of Texas at Arlington, TX, USA

SKILLS

Cloud & AI Platforms:
- AWS (S3, Lambda, Glue, EMR, Redshift, Kinesis, Step Functions, SageMaker, Bedrock, EKS, CloudWatch), Azure (Databricks, Data Factory, Synapse), Snowflake, Databricks, Delta Lake, Unity Catalog

Data & ML Engineering:
- Apache Spark, PySpark, Spark SQL, Apache Airflow, Apache Kafka, dbt, AWS Glue, Azure Data Factory, Debezium CDC, AWS DMS, MLflow, Feature Store, Vector Databases (Pinecone, FAISS), LangChain, RAG Pipelines

Languages & Frameworks:
- Python, SQL, TypeScript, PySpark, FastAPI, Pandas, Pydantic, Great Expectations, scikit-learn, PyTorch, Hugging Face Transformers

Warehousing & Modeling:
- Snowflake, Redshift, Star/Snowflake Schemas, SCD Type 1 & 2, Conformed Dimensions, Fact-Grain Design, Materialized Views

DevOps & Observability:
- Docker, Kubernetes (EKS), Terraform, GitHub Actions, Jenkins, Azure DevOps, Prometheus, Grafana, CloudWatch, OpenLineage, PagerDuty

Databases & BI:
- PostgreSQL, Oracle, SQL Server, MySQL, Power BI (DAX, RLS), Tableau

CERTIFICATIONS

- AWS Certified Data Engineer – Associate | Databricks Certified Data Engineer Associate | SnowPro Core | dbt Fundamentals

PROFESSIONAL EXPERIENCE

Keurig Dr Pepper Inc., *Dallas, TX* *Oct 2024 – Present*
AI Data Engineer
- Building AWS-based data and AI engineering solutions powering ML and GenAI use cases across sales, supply chain, manufacturing, and consumer analytics.
- Developing petabyte-scale ETL and feature pipelines using PySpark on Databricks, AWS Glue, and Snowflake to deliver reliable, analytics-ready datasets at scale.
- Designing RAG pipelines and LLM-powered applications integrating Snowflake, S3, and vector databases (Pinecone, FAISS) with embeddings from Amazon Bedrock and Hugging Face models.
- Building a centralized feature store on Databricks for batch and real-time ML inference, ensuring training and serving consistency with point-in-time correct joins.
- Orchestrating ELT and ML training workflows with Apache Airflow, integrating S3, Kafka, microservices, and partner APIs into curated lakehouse zones with full lineage in Unity Catalog.
- Tuning Spark and Snowflake workloads through partition tuning, broadcast joins, clustering keys, and materialized views, driving substantial runtime and warehouse-cost reductions.
- Modeling dimensional data marts in Snowflake using dbt with SCD Type 2 history, conformed dimensions, and fact-grain design to support executive reporting and self-service analytics.
- Collaborating with data scientists, analysts, and business stakeholders to translate ambiguous requirements into trusted, observable, and cost-optimized data products.

Deloitte, *USA* *Jun 2020 – Dec 2022*
Data Engineer
- Designed and built end-to-end AWS ETL pipelines using S3, Glue, Lambda, Step Functions, and Redshift to move client data from mainframe, Oracle, and SQL Server systems into analytics-ready datasets for enterprise reporting and regulatory submissions.
- Implemented Apache Airflow as the orchestration backbone for hundreds of DAGs, introducing standardized patterns, exponential-backoff retries, PagerDuty alerting, and SLA dashboards that materially improved pipeline reliability.
- Built data-validation frameworks using Great Expectations and custom Python checks integrated into Airflow tasks, catching schema drift, null spikes, and referential-integrity violations before downstream consumers were affected.
- Tuned PySpark and Snowflake workloads through partition tuning, broadcast hints, clustering keys, and materialized views, cutting runtime and warehouse cost on the most expensive nightly jobs.
- Designed Redshift and Snowflake warehouse models with star schemas, SCD Type 1 and Type 2, conformed dimensions, and late-arriving dimensions, supporting enterprise reporting and self-service BI for hundreds of analysts.
- Built customer-segmentation and propensity datasets including next-best-action and churn risk, enabling targeted campaigns and meaningful uplift on retention and cross-sell metrics; partnered with cross-functional teams across business, data science, and engineering to improve data accessibility and workflow efficiency.